

Normalize a Copied Inertial Record

Make evidence stable before asking what it means

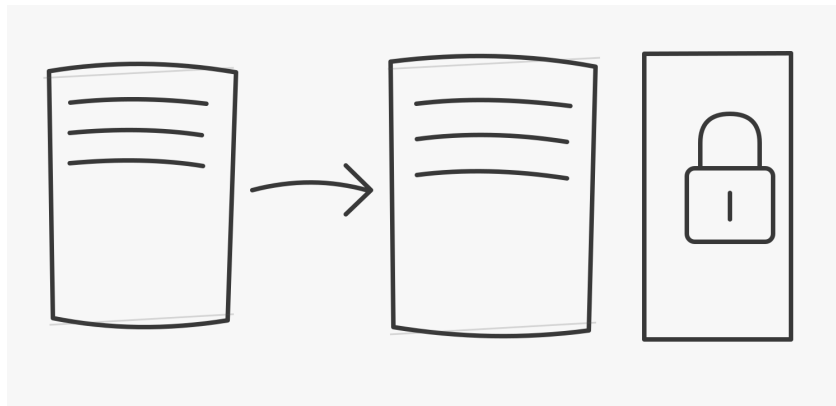
E0 SYNTHETIC REPLAY — NO SENSOR, BUS, PIN, OR CALIBRATION CLAIM

Lesson	067 — component	Time	150–210 minutes
Prerequisite	Lesson 043 copied inertial samples	Board	compile-only Mega replay
Input	one copied sample	Observation	named record and 64-byte image

Responsibility. `InertialRecordNormalizer` copies one complete Lesson 043 sample into an explicit stable record. `InertialRecordCodec` gives that record one canonical 64-byte representation. Neither object reads a sensor, rotates an axis, removes bias, qualifies a source, discovers hardware, or writes storage.

1 Draw the evidence boundary first

A measurement is useful only when its identity and conditions travel with its numbers. The copied sample already names its source, declared ranges, calibration revision, observation time, sequence, readiness, saturation, and producer status. Lesson 067 preserves that whole witness. It does not rescue a fault by borrowing vectors from the preceding healthy sample.



Alternative text: a copied sample card crosses a normalization desk into a stable typed record and fixed image; a physical sensor, bus, pins, and calibration remain behind a locked E1 gate with no connecting wire.

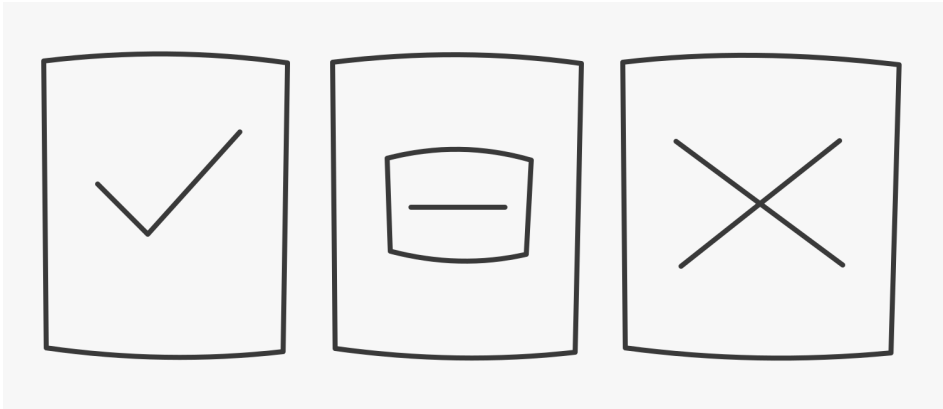
E0 establishes	E0 does not establish
field-for-field copied provenance	genuine MPU6050 or QMI8658 identity
stable revisions and closed states	physical scale, axes, bias, or accuracy
canonical bytes and corruption checks	I2C behavior, voltage, timing, or pins
deterministic replay	qualification, mounting, or interchangeability

2 Predict before normalizing

For a healthy, ready sample, predict the derived record state: _____. If the sample is healthy but not ready: _____. Write the fields that must remain visible when producer status reports a fault: _____.

3 Derive one state without discarding evidence

The record has three derived states. **Recorded** means producer status is OK and data is ready. **NotReady** means producer status is OK and data is not ready. Any non-OK producer status becomes **SourceFault**. Saturation remains a separate field; it is never rewritten as producer status.



Alternative text: three graphite cards show *Recorded* for healthy ready evidence, *NotReady* for healthy non-ready evidence, and *SourceFault* for producer failure; saturation remains visible beside every state.

Producer status	Ready	Derived state	Numeric vectors
OK	yes	Recorded	copied exactly
OK	no	NotReady	copied exactly
non-OK	either	SourceFault	copied exactly, never cached

This separation matters. “Not ready” is a valid report about an identified source; it is not the same as a transport failure. Saturation is valid evidence that a declared measurement range was exceeded; it must not be averaged away or mistaken for an ordinary maximum.

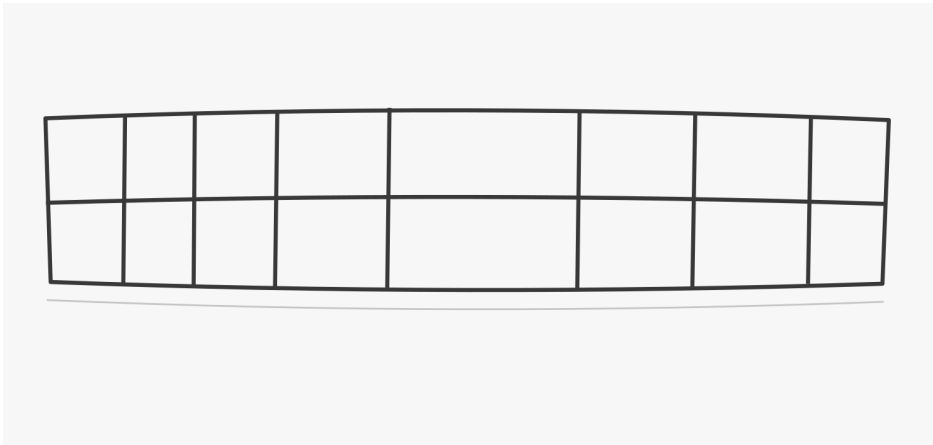
4 Keep revisions nonzero and explicit

`schemaRevision` describes the typed record contract. `normalizationRevision` describes this field-preserving transformation. Both must be nonzero. Source identity retains kind, model, source ID, source configuration revision, and calibration revision. A future lesson may compare or qualify records, but Lesson 067 merely ensures that comparison has complete inputs.

Circle each change that should produce a different canonical record: source ID; calibration revision; acceleration range; readiness; sequence; status. Explain why “the vectors are equal” is not enough: _____.

5 Lay out the canonical image deliberately

The codec writes exactly 64 bytes in little-endian order. It is a designed wire/storage-neutral image, not the in-memory layout of a C++ object. Magic, version, and encoded length establish framing. Reserved bytes are zero. A CRC-16 with polynomial `0x1021`, initial value `0xffff`, and final XOR `0x0000` covers bytes 0 through 61.



Alternative text: a hand-drawn 64-cell notebook strip groups framing, revisions, identity, ranges, two three-axis vectors, observation time, sequence, flags, status, and the final two-byte CRC seal.

Bytes	Group	Required interpretation
0–5	framing	magic, image version, encoded length
6–9	contract	schema and normalization revisions
10–19	identity	state, complete source, calibration, reserved zeroes
20–27	ranges	declared acceleration and angular-rate ranges
28–51	vectors	signed x/y/z values in Lesson 043 units
52–59	ordering	supplied modular time and sequence
60–61	quality	ready/saturation flags and producer status
62–63	integrity	little-endian CRC-16 over bytes 0–61

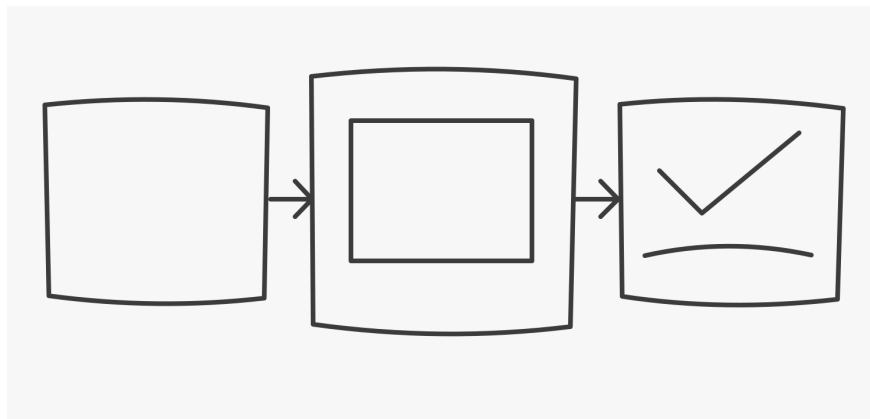
Mark the first byte range you would inspect for each symptom: wrong model _____; changed vector _____; bad CRC _____; unknown flag bit _____.

6 Do not confuse a checksum with trust

The CRC detects accidental change according to this format. It does not authenticate a producer, establish physical calibration, make volatile bytes durable, or prove that a sensor supplied them. Full typed fields remain the authoritative evidence; the CRC is an integrity check over their canonical encoding.

7 Stage the entire result before publishing

`normalize()` and codec operations are transactional. They build and validate a complete candidate first. An invalid configuration, invalid closed enum, short output span, inconsistent state, or corrupt image leaves the caller's previously published output byte-for-byte unchanged.



Alternative text: an old record and image wait beside a staging desk; only a fully checked candidate receives the pass mark and replaces both, while a failed candidate leaves the old output and its canary untouched.

1. Copy every field into a candidate typed record.
2. Derive state from producer status and readiness.
3. Validate revisions, closed enums, flags, and semantic combinations.
4. Encode all fields at locked offsets and calculate the CRC.
5. Commit the complete candidate only after every check passes.

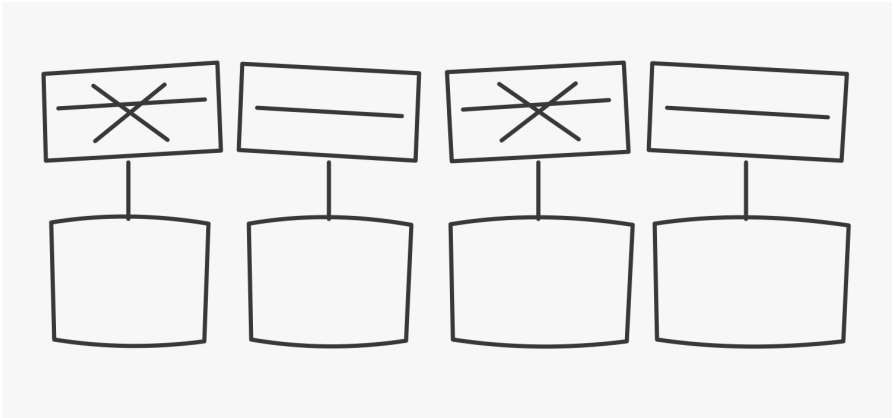
Atomicity experiment. Fill an output record and surrounding canaries with a recognizable pattern. Try each malformed sample or image. A correct implementation reports the failure and preserves the record and both canaries exactly.

8 Compare fields, not padding

Semantic record equality compares named fields. `memcmp` can observe padding, object representation, or compiler choices that are not part of the contract. Canonical image equality compares all 64 encoded bytes because that format deliberately assigns meaning to every byte, including reserved zeroes.

9 Diagnose rejection precisely

Decode first checks length, then framing, integrity, and semantic values before publishing. The typed result helps a learner locate the failed boundary without turning diagnostic text into library state.



Alternative text: four scratched record strips fall into separate diagnosis trays for *BadLength*, *BadFraming*, *BadIntegrity*, and *BadSemanticValue*; rejected strips cannot alter the prior typed output.

Validity	Example	What remains unchanged
BadLength	63 or 65 input bytes	typed output
BadFraming	magic, version, length byte, flags, reserved byte	typed output
BadIntegrity	payload changed without matching CRC	typed output
BadSemanticValue	unknown enum or contradictory state	typed output

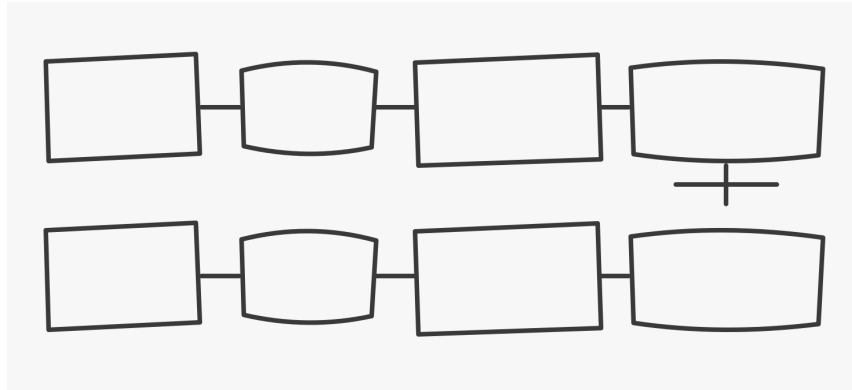
Prediction: flip one vector bit and do not update the CRC. The result should be _____. Now recompute the CRC while leaving an invalid state/ready combination. The result should be _____. This shows why integrity and meaning are independent checks.

10 Use status and validity for different questions

Status reports operation failures such as invalid configuration or insufficient destination capacity. **InertialRecordValidity** classifies an attempted decode. Neither type claims sensor health; producer health remains inside the record.

11 Replay without touching hardware

The Mega example follows acquire, configure, start, then observe, decide, actuate. It acquires one synthetic copied sample, configures two nonzero revisions, normalizes the sample, encodes the record, and exposes named volatile record and image cells. Here “actuate” means publish inert values; no endpoint is invoked.



Alternative text: two identical copied samples pass through two normalization runs and produce field-equal typed records and byte-equal 64-byte strips.

1. **Predict:** write the expected record state, identity, first vector component, sequence, flags, and image length.
2. **Observe:** compile for Mega 2560 and inspect the named volatile record and image cells with a debugger, simulator, or disassembly. Serial may support this observation but is not required.
3. **Interpret:** compare every named field and all 64 bytes across two runs. Explain any mismatch before changing the fixture.

The named copied record and image are the E0 non-Serial observation path. They establish deterministic software publication separately from resource acquisition and electrical safe state. E0 owns no resource or endpoint, so those physical checks remain open rather than silently passing.

12 Test the boundaries, not just the example

- Exercise every supported source, state, readiness, saturation, range, and producer-status class.
- Preserve negative, zero, positive, and extreme accepted components.
- Reject zero revisions, unknown enums, unknown flags, nonzero reserved bytes, inconsistent semantics, and every length error.
- Use byte-exact golden images, round trips, individual field corruption, CRC corruption, and output canaries.
- Prove that adding an unrelated later library component does not change Lesson 067 records or its resource evidence.

13 Draw the line around the next lesson

Lesson 068 may qualify exactly one explicitly configured source and map its copied vectors through one proper axis rotation. It must consume the complete Lesson 067 record. Lesson 067 itself does no axis mapping, stationary-window check, source selection, voting, failover, or pitch/roll presentation.

There is no pin-by-pin connection list, wiring diagram, or formal schematic at E0 because there is no circuit. A future E1 adapter first needs an exact chip and carrier revision, primary data sheet and register map, identity and reset behavior, address straps, byte order, ranges and scaling, data-ready behavior, mounting axes, regulator and pull-ups, logic levels, current, bus ownership, rollback, safe state, and bench acceptance. Evidence for one device family cannot close another family's card.

14 Troubleshooting matrix

Symptom	Inspect first	Interpretation
NotReady	producer status and ready flag	valid copied evidence; no new ready sample
SourceFault	complete producer status	source reported failure; vectors are not rescued
saturation set	declared range and saturation field	quality warning remains independent of state
BadFraming	magic/version/length/flags/reserved	image cannot enter the semantic boundary
BadIntegrity	bytes 0-61 and stored CRC	canonical bytes changed
BadSemanticValue	closed enums and state consistency	intact bytes encode an invalid meaning

15 Exercises

1. Explain why a fault record must preserve its current vectors rather than repeat the last healthy vectors.
2. Hand-encode one negative 32-bit vector component in little-endian order.
3. List three edits that change the CRC but not the image length.
4. Design a changed-duplicate test in which time and sequence are equal but one identity field differs.
5. Explain why an intact CRC is neither authentication nor physical qualification.

16 Open acceptance record

Host evidence	Pending integration: strict host tests, no-exception tests, standalone header, golden images, corruption matrix, atomicity, and canaries.
Mega evidence	Pending integration: compile the canonical synthetic replay and record exact flash and static SRAM.
Memory evidence	Planning limits are a 4-byte normalizer and an exact 64-byte canonical image; record measured object and synchronous-stack evidence before publication.
Document evidence	Build twice; check metadata, embedded fonts, extraction, monochrome output, links, classifications, and rendered pencil legibility.
Physical evidence	Open E1: no sensor, bus, pin, mounting, range, calibration, data-ready, safe-state, or bench observation has been established.

Acceptance item	Record
Commit tested	
Toolchain and Mega core	
Host and sanitizer result	
Flash / static SRAM / stack	
Record object / normalizer / image sizes	
Golden-image and corruption result	
PDF visual and extraction review	
Open E1 owner and next action	

Stop condition. Do not call this hardware verified. The completed E0 exercise proves a deterministic record contract over copied values. Exact powered acquisition, physical axes, calibration, observation, and safe state remain separate work.